MEANS TOOLBOX

t procedures • new 4.1–4.10 • old 7.1–7.10

ZONE A · ONE MEAN, PAIRED DIFFERENCES, TWO MEANS

ONE MEAN

$$s_{\bar{x}} = \frac{s}{\sqrt{n}}$$

$$\bar{x} \pm t^* \frac{s}{\sqrt{n}} \qquad df = n - 1$$

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \qquad df = n - 1$$

$H_0 : \mu = \mu_0$. Standard error $s_{\bar{x}}$ estimates $\sigma_{\bar{x}} = \sigma/\sqrt{n}$ using sample SD s.

PAIRED · WORK WITH THE DIFFERENCES

Same subjects measured twice, or natural pairs: subtract in one consistent order.

$\bar{d}$: mean difference; s_d: SD of differences; n: number of **pairs**.

$$t = \frac{\bar{d} - \mu_{d0}}{s_d/\sqrt{n}} \qquad df = n - 1$$

Usually $H_0 : \mu_d = 0$. Treat the differences as **one sample**; use their shape and their SD.

TWO MEANS · INDEPENDENT GROUPS

$$s_{\bar{x}_1 - \bar{x}_2} = \sqrt{s_1^2/n_1 + s_2^2/n_2}$$

$$\sigma_{\bar{x}_1 - \bar{x}_2} = \sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}$$

Left: SE estimates the sampling SD. Right: true SD when population SDs σ_1, σ_2 are known.

$$(\bar{x}_1 - \bar{x}_2) \pm t^* \sqrt{s_1^2/n_1 + s_2^2/n_2}$$

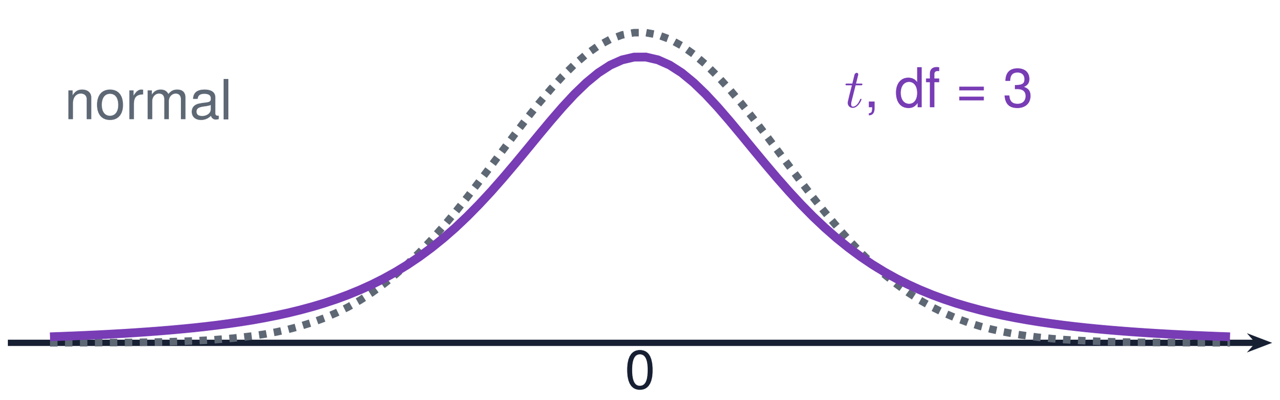
df: calculator, or min(n_1-1, n_2-1)

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$$

df: calculator, or min(n_1-1, n_2-1)

Test above assumes $H_0 : \mu_1 = \mu_2$. Calculator df is Welch's approximation; the minimum is conservative.

ZONE B · WHY t? + CONDITIONS IN ORDER



t because we replaced σ with s.

Heavier tails; more df $\longrightarrow$ closer to normal.

1 · Random: random sample or randomized experiment.
2 · 10%: $n < 0.10N$ when sampling without replacement.
3 · Normal / Large Sample: normal population, $n \geq 30$, or graph shows no strong skew or outliers.

Paired: check the **differences**.

Two samples: independent groups; check **each group**.

ZONE C · SAY IT IN WORDS

"We are ____% confident the true mean ________________ is between _____ and _____ _____ [units]."

For paired data, name the mean difference μ_d; for independent groups, name $\mu_1 - \mu_2$.

ZONE D · WATCH OUT

Paired or two-sample? Let the study design decide. Repeated measurements or natural pairs use differences; separate independent groups use two-sample t.

Keep group variances separate for two-sample t. Use the unpooled SE above.

see WS old u7_l1–10